# ATM - Atomic Physics

## Signature Sheet

Student's Name _____ Partner's Name _____

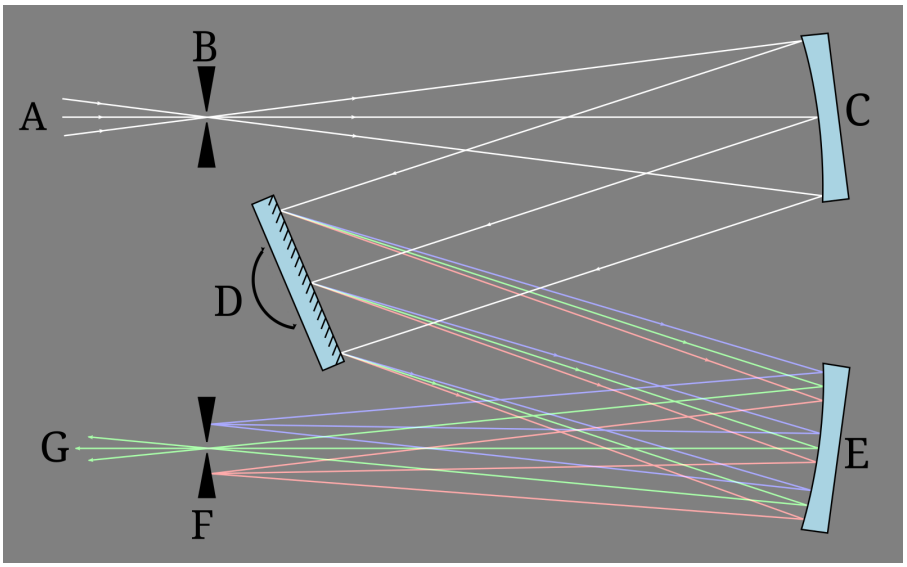
**Suggested reading to start with:**

1. Griffith, "**Introduction to Quantum Mechanics**":
   1) Chapter 4.2: Hydrogen Atom
   2) Chapter 5.1 & 5.2: Two Partical System (Helium)
   3) Chapter 7.4: Zeeman Effect
2. Thorlabs, "**Diffraction Gratings Tutorial**"
3. Thorlabs, "**Fabry–Perot Interferometer Tutorial**"

**Pre-Lab Discussion Questions**

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. Draw two diagrams before you answer your Pre-Lab Discussion Questions: (1) the energy levels of hydrogen with the transitions of the Balmer series; (2) the singlet and triplet energy levels of helium, with the transitions that give the red and yellow lines — then, on the two levels of the red line, indicate the Zeeman splitting when a magnetic field is applied. Answer the following questions based on your diagrams and correct/polish your diagrams through preparing these questions.
2. What is the formula that gives the wavelengths of these lines for the simple Bohr hydrogen atom? What is the definition of Rydberg constant in this expression? How does the derivation of the energy levels in the Bohr model differ from the quantum mechanical methods and how quantum mechanics express **Rydberg constant** in Hydrogen model?
3. Write the following atomic state's abbreviation in "full spectroscopic notation" (the notation for states written in the $n,l,j,m_j$ basis, described in the first instructional video).A hydrogen atom is in the n=3, l=2 state. Note that there are 2 possible values for J. What do these quantum numbers mean?
4. Show the transitions that produce the red and yellow lines in the Helium diagram and explain what is singlet and triplet. What is the main difference of Helium model comparing to Hydrogen model?
5. When a 1 Tesla magnetic field is applied to helium, what happens to the energy levels and transitions that produce the red line of helium? Now give the physical meaning of **Bohr magneton**.
6. Explain how the grating works. Can we use zeroth diffraction order to separate lights of different colors? Would higher order diffraction orders affect the measurements? Calculate a representative value for the resolving power of the grating with given period/grating constant. Then, based on the figure below, explain how our monochromator uses the grating to select a single wavelength for the exit slit.




7. What gives a spectral line a non-zero width? Estimate the line width for the first Balmer series. Assume that the pressure inside the tube is 5 torr and the temperature 600K, and that the lifetimes of the states are about 10ns. (hint: consider the resolution of the spectrum, Doppler broadening and the time uncertainty principle, which one is the main factor?)
8. Briefly explain how FB interferometer works. Calculate the resolving power of the interferometer. Why is it necessary to use the interferometer instead of the grating for observing the Zeeman Effect. Can we observe Zeeman effect through our monochromator?

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

**Mid-Lab Discussion Questions**

1. On day 3 of this lab, you should have successfully produced a plot of the Balmer-series lines, and made an estimate of the Rydberg constant. Show them to an instructor, briefly explain how you corrected the result and ask for a signature.

Staff Signature _____ Date _____

Completed by day 3 of lab? (Circle one) Yes / No

1. On day 5 of this lab, you should have successfully observed the Zeeman splitting of the helium lines and estimated a value for the Bohr magneton. Demonstrate this to an instructor, explain the strategy you are using to measure the Bohr magneton and ask for a signature.

Staff Signature _____ Date _____

Completed by day 5 of lab? (Circle one) Yes / No

**Checkpoint Signatures**

1. Preparation

   Staff Signature _____

2. Peak Finding

   Staff Signature _____

3. Additional Questions

   Staff Signature _____

4. Zeeman Picture

   Staff Signature _____

5. Zeeman Splitting

   Staff Signature _____

**Station Clean-Up Check-Off**

Before leaving, ensure your station is clean and ready for the next group. The GSI must verify and sign off on the following:

1. Any BNC cables that you have been moving are unplugged; return things to the state that you found them.
2. AFM prepared samples are removed and the sample disks are returned to the bag. CD/DVD/Blu-Ray scraps are thrown out if small, put in the bin if large.
3. All used pipette tips go into trash and glass sample slides go into glass disposal container; pipettes are put back to the shelf.
4. Workspace cleared of personal belongings, papers, and debris.
5. Any spills or hazards cleaned up; no left-over onion in the fridge or on the bench.
6. Microscope is clear of mounted sample, turned off and covered by the plastic bag properly if applicable.
7. Restart the computer at your station so that any software updates are applied.
8. Station is clean and ready for the next group. Staff discretion for additional cleanup that is necessary applies.

Staff Signature _____ Date _____